PHYSICS

Standard Level

Set A | Paper 2

Challenge mock examination

Time allowed: 90 minutes Maximum mark: 50

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• All structured questions are compulsory. There are no optional topics or optional questions.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

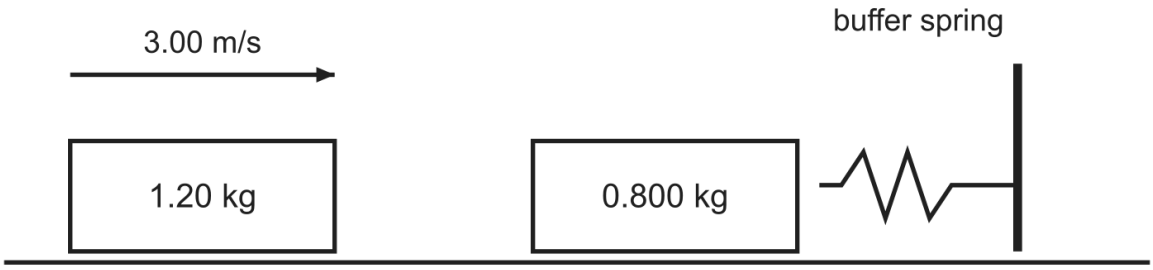
Thriving Scholars challenge practice | Current Physics syllabus | 2026

2 Question 1 | Question paper

Paper 2 - Structured questions

2 Question 1 [8 marks]

A $1.20\,\text{kg}$ cart moves at $3.00\,\text{m}\,\text{s}^{-1}$ and couples to a stationary $0.800\,\text{kg}$ cart on a horizontal track. External impulse during the $0.0500\,\text{s}$ coupling is negligible. The coupled carts then compress an initially unstretched ideal spring of constant $180\,\text{N}\,\text{m}^{-1}$. Resistance after the collision is negligible.



(a) Calculate the common velocity immediately after coupling. [2]

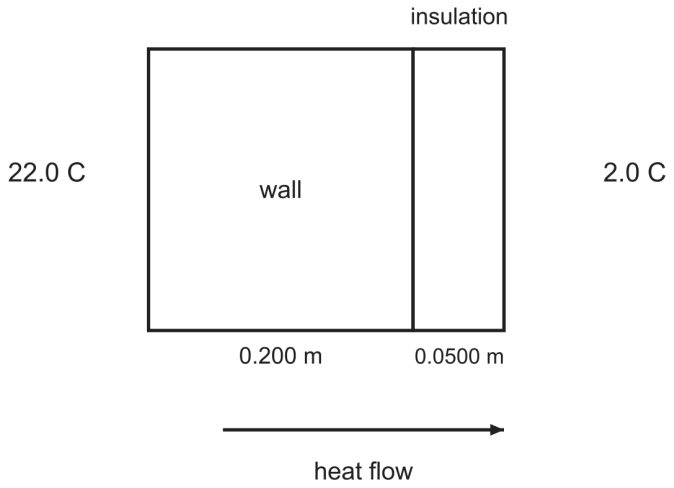
(b) Calculate the average force exerted on the initially stationary cart during coupling. [2]

(c) Determine the fraction of the initial total kinetic energy dissipated during coupling. [2]

(d) Calculate the maximum spring compression. [2]

2 Question 2 [8 marks]

A wall of area $12.0\,\mathrm{m}^2$, thickness $0.200\,\mathrm{m}$ and thermal conductivity $0.800\,\mathrm{W\,m^{-1}\,K^{-1}}$ separates air at $22.0^\circ\mathrm{C}$ from air at $2.0^\circ\mathrm{C}$. Treat the wall surfaces as being at these temperatures. An insulating layer of thickness $0.0500\,\mathrm{m}$ and conductivity $0.0400\,\mathrm{W\,m^{-1}\,K^{-1}}$ is added to the cold side. Assume steady one-dimensional conduction and no other losses. The heater converts 80.0% of electrical input into useful room heating.



(a) Describe conduction in the solid wall at the particle level. [2]

(b) Calculate the heat-transfer rate through the bare wall. [2]

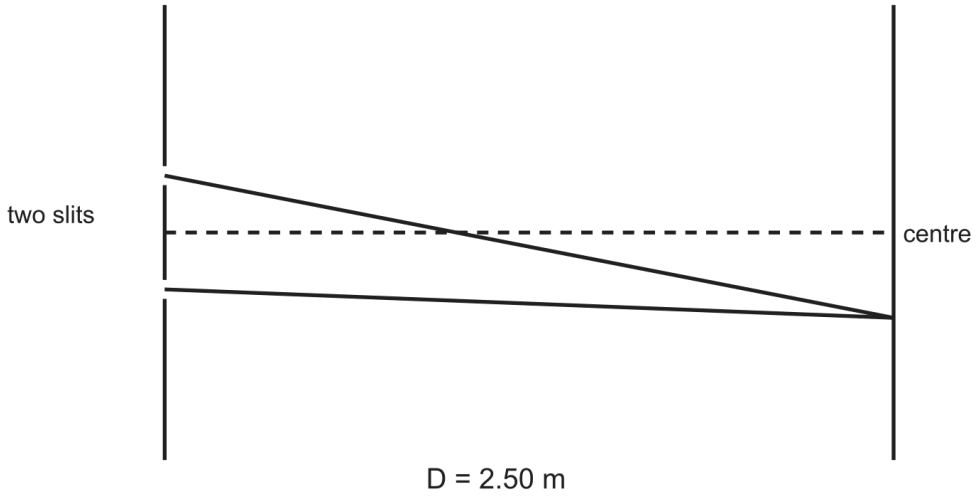
(c) Calculate the heat-transfer rate after insulation is added. [2]

(d) Calculate the wall-insulation interface temperature. [1]

(e) Calculate the electrical energy saved in $10.0\,\mathrm{h}$, in kWh. [1]

2 Question 3 [8 marks]

Coherent light of wavelength $600\,\text{nm}$ in air illuminates two narrow slits separated by $0.300\,\text{mm}$. A screen is $2.50\,\text{m}$ from the slits. Use the small-angle approximation. The light frequency remains fixed throughout.



(a) Explain why a fixed phase relationship between the slits is required for a stable fringe pattern. [2]

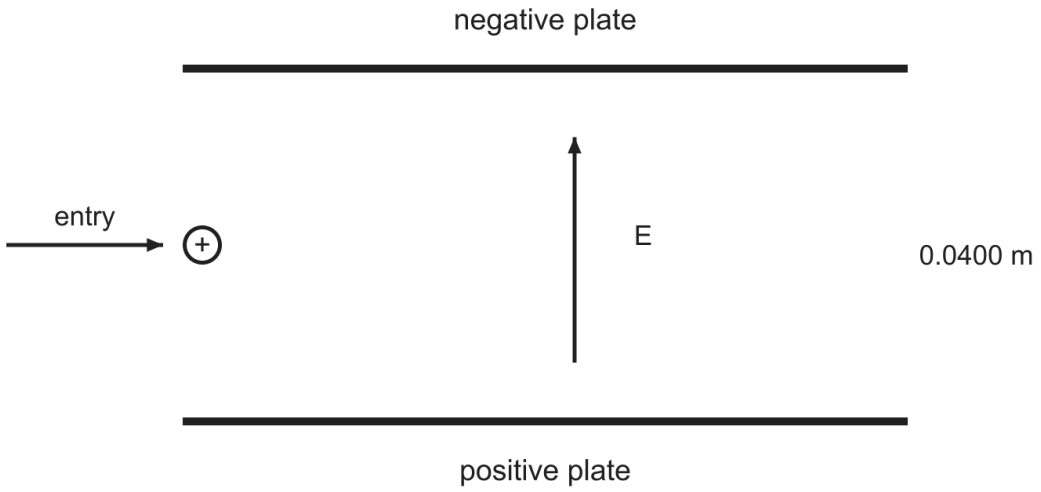
(b) Calculate the separation of adjacent bright fringes. [2]

(c) Calculate the distance between the first dark fringe to the left of the central maximum and the second dark fringe to its right. [2]

(d) The entire slit-screen region is filled with a transparent medium of refractive index 1.33. Calculate the new bright-fringe separation and explain the change. [2]

2 Question 4 [8 marks]

A proton enters midway between horizontal parallel plates with horizontal speed $1.00 \times 10^{5}\ \mathrm{m\,s^{-1}}$. Plate separation is $0.0400\ \mathrm{m}$, plate length $0.0300\ \mathrm{m}$ and potential difference $240\ \mathrm{V}$. The upper plate is negative. Neglect gravity and edge effects. Use proton charge $1.60 \times 10^{-19}\ \mathrm{C}$ and mass $1.67 \times 10^{-27}\ \mathrm{kg}$.



(a) Calculate the electric-field strength and the force on the proton, including direction. [2]

(b) Calculate the proton acceleration. [2]

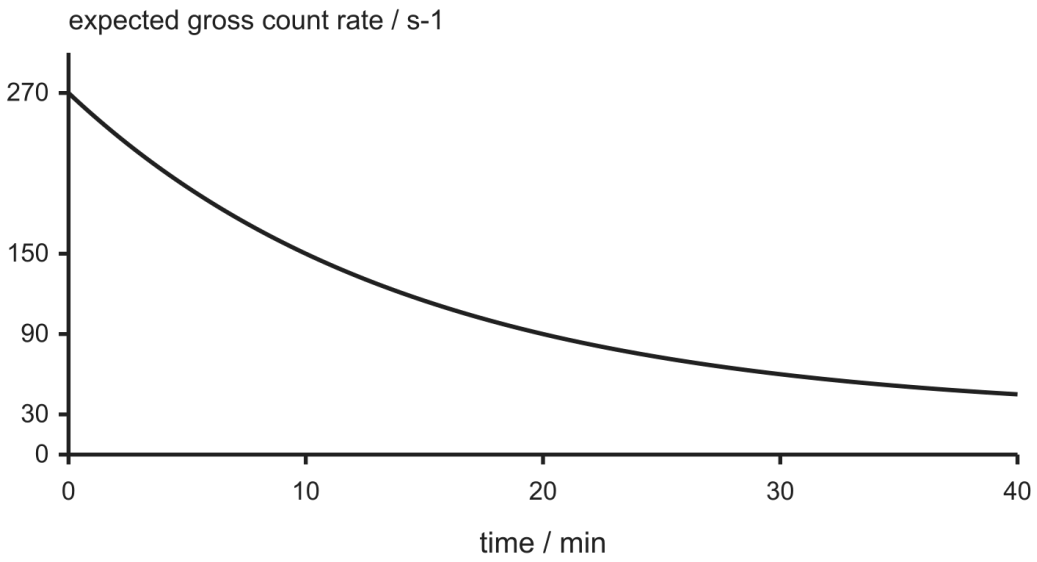
(c) Assuming it could travel the full plate length, calculate its vertical displacement by the exit plane. [2]

(d) Decide whether the proton actually reaches that plane while remaining between the plates. Justify your conclusion. [2]

2 Question 5 | Question paper

2 Question 5 [9 marks]

An activated sample undergoes beta-minus decay, forming a stable daughter. Its initial parent-nucleus count is 8.00×10^{9}. A detector has a constant background rate $30.0\,\mathrm{s}^{-1}$. Its expected gross count rates are $270\,\mathrm{s}^{-1}$ at $t = 0$ and $90.0\,\mathrm{s}^{-1}$ at $t = 20.0\,\mathrm{min}$. Detector efficiency remains constant. The illustrative curve shows expected rates rather than individual random measurements.



(a) For a parent ${}^{A}_{Z}X$, write the beta-minus decay equation with a daughter Y and all emitted particles. [2]

(b) Determine the sample half-life. [2]

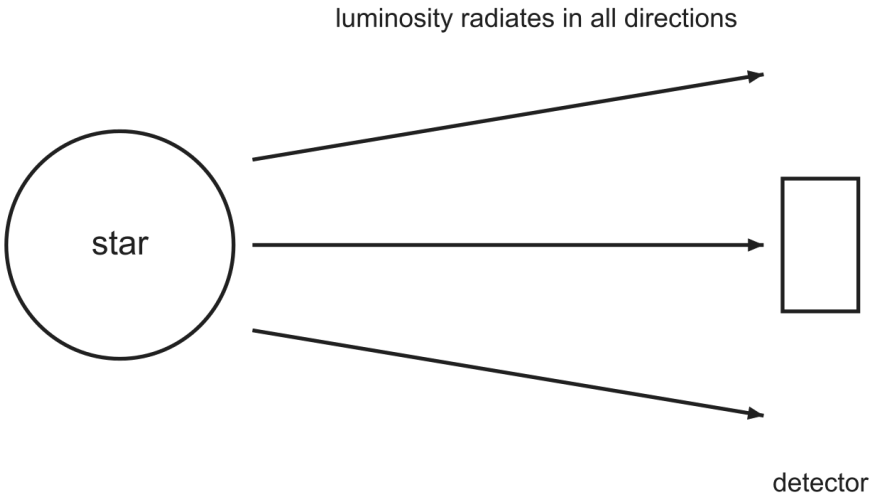
(c) Predict the gross count rate at $30.0\,\mathrm{min}$. [2]

(d) Determine the number of parent nuclei remaining at $30.0\,\mathrm{min}$. [1]

(e) Explain why a short measured count rate need not lie exactly on the curve and can occasionally be below the expected background rate. [2]

2 Question 6 [9 marks]

A star has luminosity $3.00 \times 10^{27}\,\text{W}$, apparent brightness $2.50 \times 10^{-10}\,\text{W m}^{-2}$ and surface temperature $6000\,\text{K}$. Ignore interstellar absorption and treat the surface as a black body. Use $\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$, Wien constant $2.90 \times 10^{-3}\,\text{m K}$ and $1\,\text{pc} = 3.086 \times 10^{16}\,\text{m}$.



(a) Calculate the distance to the star in metres. [2]

(b) Calculate its annual parallax angle in arc-seconds. [2]

(c) Calculate the star radius. [2]

(d) Calculate its peak-emission wavelength. [1]

(e) Explain how a main-sequence star can maintain a nearly constant radius while nuclear fusion continues. [2]

End of question paper

PHYSICS

Standard Level

Set A | Paper 2

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 2 - Worked solutions

2 Question 1 [8 marks]

(a) [2 marks]

M1 $1.20(3.00) = (1.20 + 0.800)v$ by momentum conservation.

A1 $v = 1.80\,\text{m}\,\text{s}^{-1}$ in the initial direction.

(b) [2 marks]

M1 $F_{\text{av}} = \Delta p/\Delta t = 0.800(1.80)/0.0500$.

A1 $F_{\text{av}} = 28.8\,\text{N}$ in the initial moving cart's direction.

(c) [2 marks]

M1 $K_i = \frac{1}{2}(1.20)(3.00)^2 = 5.40\,\text{J}$ and $K_f = \frac{1}{2}(2.00)(1.80)^2 = 3.24\,\text{J}$.

A1 The dissipated fraction is $(5.40 - 3.24)/5.40 = 0.400$.

(d) [2 marks]

M1 After coupling, $\frac{1}{2}kx^2 = 3.24\,\text{J}$. Do not reuse the pre-collision kinetic energy.

A1 $x = \sqrt{2(3.24)/180} = 0.190\,\text{m}$.

2 Question 2 [8 marks]

(a) [2 marks]

R1 Particles at the hot side have greater mean random vibrational energy and transfer energy to neighbouring particles through interactions.

R1 There is net energy transfer down the temperature gradient without bulk motion of the solid; free electrons can also carry energy in a conducting material.

(b) [2 marks]

M1 $P = kA\Delta T/L = 0.800(12.0)(20.0)/0.200$.

A1 $P = 960\,\text{W}$.

(c) [2 marks]

M1 Add series thermal resistances: $P = A\Delta T/(L_1/k_1 + L_2/k_2)$.

A1 $P = 12.0(20.0)/(0.200/0.800 + 0.0500/0.0400) = 160\,\text{W}$.

(d) [1 mark]

A1 The wall drop is $PL_1/(k_1A) = 3.33\,\text{K}$, so the interface is $22.0 - 3.33 = 18.7^\circ\text{C}$.

(e) [1 mark]

A1 Input power saved is $(960 - 160)/0.800 = 1000\,\text{W} = 1.00\,\text{kW}$, so energy saved is $10.0\,\text{kWh}$.

2 Question 3 [8 marks]

(a) [2 marks]

R1 A stable phase difference lets the same positions retain constructive or destructive interference.

R1 If the phase difference changes randomly, bright and dark positions move rapidly and the time-averaged pattern washes out.

(b) [2 marks]

M1 $s = \lambda D/d = (600 \times 10^{-9})(2.50)/(0.300 \times 10^{-3})$.

A1 $s = 5.00\,\text{mm}$.

(c) [2 marks]

M1 Dark positions are at half-integer fringe spacings. The requested positions are $-0.5s$ and $+1.5s$.

A1 The separation is $2s = 10.0\,\text{mm}$.

(d) [2 marks]

M1 Frequency is unchanged while speed and wavelength are reduced by factor 1.33, so fringe spacing is also divided by 1.33.

A1 $s_{\text{new}} = 5.00/1.33 = 3.76\,\text{mm}$.

2 Question 4 [8 marks]

(a) [2 marks]

A1 $E = V/d = 240/0.0400 = 6.00 \times 10^{3}\ \text{V m}^{-1}$, upward toward the negative plate.

A1 $F = qE = 9.60 \times 10^{-16}\ \text{N}$, upward.

(b) [2 marks]

M1 $a = F/m = 9.60 \times 10^{-16} / (1.67 \times 10^{-27})$.

A1 $a = 5.75 \times 10^{11}\ \text{m s}^{-2}$, upward.

(c) [2 marks]

M1 Horizontal speed is unchanged, so $t = L/v_x = 3.00 \times 10^{-7}\ \text{s}$.

A1 $y = \tfrac{1}{2}at^2 = 0.0259\ \text{m}$.

(d) [2 marks]

R1 The available upward clearance from the entry midpoint is only $0.0200\ \text{m}$, less than the projected displacement.

R1 The proton therefore hits the upper plate before the exit plane; the unimpeded trajectory used in (c) cannot describe motion beyond contact. Accept the equivalent calculation $x_{\text{hit}} = 0.0264\ \text{m} < 0.0300\ \text{m}$.

2 Question 5 [9 marks]

(a) [2 marks]

A1 The daughter has nucleon number A and proton number $Z + 1$.

A1 ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + \bar{\nu}_e$. An electron antineutrino accompanies the beta-minus electron.

(b) [2 marks]

M1 Subtract background: source rates are 240 and $60.0\,\text{s}^{-1}$, a reduction by 4.

A1 This is two half-lives in 20.0 min, so $T_{1/2} = 10.0\,\text{min}$.

(c) [2 marks]

M1 After three half-lives the source rate is $240/8 = 30.0\,\text{s}^{-1}$.

A1 Add background to obtain $60.0\,\text{s}^{-1}$ gross.

(d) [1 mark]

A1 $N = 8.00 \times 10^{9}/8 = 1.00 \times 10^{9}$ parent nuclei.

(e) [2 marks]

R1 Radioactive detection and background events are random, so finite-interval counts fluctuate around their expected values.

R1 The expected background is a mean, not a strict lower bound on a measured short-interval count rate; sufficiently long averages approach the expected curve. No negative physical source rate is implied.

2 Question 6 [9 marks]

(a) [2 marks]

M1 $b = L/(4\pi d^2)$, so $d = \sqrt{L/(4\pi b)}$.

A1 $d = 9.77 \times 10^{17}\,\text{m}$.

(b) [2 marks]

M1 $d = 31.7\,\text{pc}$ using the given conversion.

A1 $p = 1/d_{\text{pc}} = 0.0316$ arc-seconds.

(c) [2 marks]

M1 $L = 4\pi R^2 \sigma T^4$, so $R = \sqrt{L/(4\pi\sigma T^4)}$.

A1 $R = 1.80 \times 10^9\,\text{m}$.

(d) [1 mark]

A1 $\lambda_{\text{max}} = 2.90 \times 10^{-3}/6000 = 4.83 \times 10^{-7}\,\text{m}$ or $483\,\text{nm}$.

(e) [2 marks]

R1 Inward gravitational attraction is balanced by an outward pressure gradient in hydrostatic equilibrium.

R1 Energy from core fusion helps maintain the temperature and pressure; energy transported outward is radiated from the surface, allowing approximate equilibrium over main-sequence timescales.

End of worked solution book